
BACKGROUND

- 2024 – **Assistant Professor**, *James Madison University*, Department of Mathematics and Statistics, Harrisonburg, VA, USA
- 2022 – 2024 **Postdoctoral Staff Scientist (Wissenschaftlicher Mitarbeiter)**, *Interactive Optimization and Learning (IOL) Lab* , *Technische Universität Berlin (TU Berlin)* & *Zuse Institute Berlin (ZIB)*, Berlin, Germany
- Advisor: Sebastian Pokutta
- 2021 **Ph.D., Mathematics**, *North Carolina State University*, Raleigh, NC, USA
- Thesis: **Construction of Functions from Nonlinear Transformations**
- Advisor: Patrick L. Combettes
- 2018 **M.Sc., Applied Mathematics**, *North Carolina State University*, Raleigh, NC, USA
- 2016 **B.Sc., Mathematics** (Summa cum Laude), *James Madison University*, Harrisonburg, VA, USA

AWARDS, HONORS, & RECOGNITIONS

- 2025 – **Principal Investigator (NSF DMS-2532423)**, *National Science Foundation*
“LEAPS-MPS: Modernizing Frank-Wolfe splitting algorithms”  (224,697 USD)
- 2023 – 2025 **Principal Investigator (EF1-23)**, *Berlin Mathematics Research Center MATH+ Grant*
“On a Frank-Wolfe Approach for Abs-smooth Optimization”  (255,000 EUR), PIs with S. Pokutta and A. Walther.
- 2022 – 2024 **Research Group Lead (Continuous Optimization)**, *IOL Lab*, *TU Berlin* & *ZIB*
2022 **MATH+ Travel Grant**, *Berlin Mathematics Research Center*
- 2022 & 2023 **REU Advisor**, *Zuse Institute Berlin*
Advising on research, acquiring visas and funding, general organization
- 2022 – 2024 **MATH+ Postdoctoral Member**, *Berlin Mathematics Research Center*
- 2018 – 2021 **Graduate Research Fellowship (NSF-GRFP DGE-1746939)**, *National Science Foundation*, at NC State University, “Convolutional Methods for Data Science” (138,000 USD)
- 2016 **University Graduate Fellowship**, *NC State University*
- 2016 **Ikenberry Prize**, *James Madison University*, Department of Mathematics & Statistics
Presented to the outstanding member of the senior class
- 2015 **Joan and Ernest Droms Memorial Scholarship in Mathematics**, *James Madison University*,
- 2014 & 2016 **COMAP International Mathematical Contest in Modeling (MCM)**
2016: *Meritorious* (top 8%); 2014: *Honorable Mention*
- 2014 **Jeffrey E. Tickle Scholarship in Mathematics**, *James Madison University*
- 2013 & 2014 **Lisa Persson-Helms Scholarship in Mathematics**, *James Madison University*
- 2014 **Putnam Exam**, *James Madison University*, Score: 9

RESEARCH

Current research topics

- (1) Design and convergence analysis for split and block-iterative first-order continuous optimization algorithms, (2) Analysis for the estimation of Hoffman constants for nonlinear systems of closed/convex sets, (3) Developing algorithms for use in signal recovery/synthesis, robotics, and biological applications, and (4) Nonlinear analysis and fixed-point relaxations of optimization problems.

PROJECTS

- 2025 – **Optimization & Algorithmic Theory (OAT) Lab**, *James Madison University (Dept. of Math & Stats)*
Undergraduate optimization lab with focus on the development, analysis, and applications of first-order continuous optimization algorithms. <https://zevwoodstock.github.io/oatlab/> 
- 2022 – 2024 **Interactive Optimization and Learning (IOL) Lab**, *TU Berlin (Math Dept.)* & *Zuse Institute Berlin (AI in Society, Science, and Technology Dept.)*, Advisor: Sebastian Pokutta
Research on continuous optimization with applications in machine learning and data science.
- 2017 – 2021 **Construction of Functions from Nonlinear Transformations**, *NCSU*, Advisor: Patrick L. Combettes
PhD Thesis: modeling, analysis, and algorithm development for solving nonlinear problems in signal/image/audio processing, data science, and large-scale optimization.

- 2014 – 2016 **Independent Study: Aeroacoustics of Turbulent Coanda Wall Jets**, *JMU*, Advisor: Caroline Lubert
 2015 **NSF-REU** (Algebraic methods in computational biology), *Texas A&M University*, Advisor: Anne Shiu
 2014 **NSF-REU** (Inverse scattering problems), *James Madison University*, Advisor: Hala Nelson

TECHNICAL

High-level languages, *MATLAB/Octave, Julia, Python 3, and FORTRAN*

Markup languages, *L^AT_EX, Beamer, Microsoft Office, Markdown, and HTML*

Tools, *Linux/Mac/Windows, bash, git, docker, slurm (HPC), ssh, vim, meld, Canva, LLMs, Maple, Mathematica, Canvas, and Moodle*

Programming Experience

Numerical work in articles 3 – 16; new algorithms in articles 5, 6, 9–12, 15–16. Coded projects for optimization (convex, nonconvex, asynchronous, parallelized, block-iterative, nonsmooth, ...), signal/image/audio processing, nonlinear fixed point equations, control theory, finite element methods, numerical integration/differentiation, computational geometry, classification, image recognition, encoder/decoder models.

PEER-REVIEWED ARTICLES

(Undergraduate advisees indicated by *)

ARTICLES UNDER REVIEW

16. **Frank-Wolfe with Moreau envelope smoothing for nonsmooth nonconvex problems** [✉](#), Antonio Silveti-Falls, Cesare Molinari, and Z. W., submitted (2026+).
15. **A conditional-gradient-based single-loop augmented Lagrangian method for inequality constrained optimization** [✉](#), Xiaozhou Wang, Ting Kei Pong, and Z.W., submitted (2026+).
14. **A note on asynchronous Projective Splitting in Julia** [✉](#), Utkarsh Sharma*, Kashish Goel*, Aryan Dua*, Sebastian Pokutta, and Z.W., submitted (2025+).

PUBLISHED ARTICLES

13. **Flexible block-iterative analysis for the Frank-Wolfe algorithm** [✉](#), G. Braun, J. Halbey, S. Pokutta, and Z.W., **Journal of Optimization Theory and Applications**, vol. 210, no. 37 (2026).
12. **High-precision linear minimization is no slower than projection** [✉](#), Z.W., **Optimization Letters**, vol. 20, pp. 909–915 (2026).
11. **Splitting the conditional gradient algorithm** [✉](#), Z. W. and S. Pokutta, **SIAM Journal on Optimization**, vol. 35, iss. 1, pp. 347–368, (2025).
10. **On a Frank-Wolfe approach for abs-smooth functions** [✉](#), T. Kreimeier, S. Pokutta, A. Walther, and Z. W., **Optimization Methods and Software**, pp. 1–27, (2024).
9. **Signal recovery from inconsistent nonlinear observations** [✉](#), P. L. Combettes, and Z. W., **Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)**, pp. 5872–5876. Singapore, May 22–27, (2022).
8. **Block-activated algorithms for multicomponent fully nonsmooth minimization** [✉](#), M. N. Bui, P. L. Combettes, and Z. W., **Proceedings of the 2022 IEEE ICASSP**, pp. 5428–5432. Singapore, May 22–27, (2022).
7. **A variational inequality model for the construction of signals from inconsistent nonlinear equations** [✉](#), P. L. Combettes and Z. W., **SIAM Journal on Imaging Sciences**, vol. 15, pp. 84–109 (2022).
6. **Reconstruction of functions from prescribed proximal points** [✉](#), P. L. Combettes and Z. W., **Journal of Approximation Theory**, vol. 268, art. 105606, (2021).
5. **A fixed point framework for recovering signals from nonlinear transformations** [✉](#), P. L. Combettes and Z. W., **Proceedings of the European Signal Processing Conference**, pp. 2120–2124. Amsterdam, NL, January 18–22, (2021).
4. **Obstructions to convexity in neural codes** [✉](#), C. Lienkaemper*, A. Shiu, and Z. W.*, **Advances in Applied Mathematics**, vol. 85, pp. 31–59 (2017).
3. **Rocket launch noise and the Coanda effect** [✉](#), C. P. Lubert, J. N. Romero*, J. S. Sochacki, and Z. W.* **AIAA Space**, art. 2016-5625, (2016).
2. **Analyzing multistationarity in chemical reaction networks using the determinant optimization method** [✉](#), B. Félix*, A. Shiu and Z. W.*, **Applied Mathematics and Computation**, vol. 287–288, pp. 60–73 (2016).
1. **Architectural acoustical oddities.** [✉](#) Z. W.* and C. P. Lubert, **Proceedings of Meetings on Acoustics**, vol. 22, art. 025002, (2015).

TEACHING

- 2024 – **Instructor of Record**, *James Madison University*, Harrisonburg, VA, USA
Introductory Calculus (Math 205), Calculus I (Math 235), Calculus II (Math 236), Calculus III (Math 237), Computers & Numerical Algorithms (Math 248), and Modern Optimization Algorithms (Math 485 - special topics)
- 2018 – **Mentor/Advisor**, *OAT Lab* (2024 –), *IOL Lab* (2022 – 2024) and *NC State UUG* (2018 – 2021)
Advising on PhD/MS/BS research projects; acquiring grant funding; organizing seminars/trainings; reviewing applications; coaching students on curriculum, jobs, grad school, and conference presentations.

STUDENT STATISTICS:	Gender	M/F/Nonbinary	8/6/1
	Degree	PhD/MS/BS	2/2/11
	Nationality	FR/GER/USA/India	1/2/8/4

- 2022 – 2024 **ZIB Tutorial Lecture Series Organizer**, *Zuse Institute Berlin*, Berlin, Germany
Organizing and presenting expository lectures on useful topics which are not found in “standard” coursework.
- 2016 – 2018 **Recitation Leader and Teaching Assistant**, *NC State University*
Calculus I for business and life science, Calculus I for math majors and engineers, and Calculus III
- 2016 – 2018 **Math Tutor**, *NCSU Multimedia Center* (2017–2018); *JMU Science & Math Learning Center* (2016)
- 2015 – 2016 **Math 167**, *James Madison University*
Proposed, developed, and co-taught a seminar on higher-level topics in math, accessible to first-year undergraduates.

HIGHLIGHTED GRADUATE COURSEWORK

*Analysis	Nonlinear Equations and Optimization
*Numerical Analysis	Vector Space Methods in System Optimization
*Dynamical Systems and Control Theory	Real Algebraic Geometry and Convex Optimization
Nonlinear Programming	Proximal Optimization in Data Science
Convex Analysis	Representations of High-Dimensional Data (MSRI Summer School ↗)
Advanced Functional Analysis	Semidefinite and Second-Order Conic Programming

* indicates a year-long sequence for which a Ph.D. Qualifying Examination was passed.

SERVICE & EXTRACURRICULARS

COMMITTEES, LEADERSHIP, & CERTIFICATIONS

- 2024 – **Faculty Research & Awards Committee**, *JMU College of Science & Mathematics*, Harrisonburg, VA
- 2025 – **High Performance Computing (HPC) Demo + Q&A Organizer**, *JMU College of Science & Mathematics*, Harrisonburg, VA
Organized training events and resources for faculty/students to use high performance computing on campus.
- 2024 – **Applied Math Committee Member**, *JMU Department of Mathematics & Statistics*, Harrisonburg, VA
Contributed to proposal for new Computational Analytics major; co-organized redesign of the following courses: Operations Research, Modern Optimization Algorithms, and Graph Theory.
- 2026 **HPC System Administrator Search Committee**, *JMU College of Science & Mathematics*, Harrisonburg, VA
- 2025 **Excellence in STEM Teaching Program**, *JMU College of Science & Mathematics*, Harrisonburg, VA
Program on best practices for inclusive teaching for students of diverse and underrepresented backgrounds.
- 2025 **CITI Certification for Responsible Conduct of Research**
- 2022 – 2024 **Organizer**, *AI in Society, Science, and Technology (AIS²T) Seminar*, Zuse Institute Berlin, Berlin, Germany
- 2022 – 2024 **Curator**, *Award Repository*, IOL Lab, Berlin, Germany
Maintained a list of awards, reminding/nominating students, and assisted with applications.
- 2023 **Volunteer Organizer**, *Lange Nacht der Wissenschaften* (“*Long night of science*”) [↗](#), ZIB [↗](#), Berlin, Germany
- Outreach event for the Berlin science community to exhibit discoveries for the public (accessible for all ages).
- 2022 **Session Chair**, *Nonsmooth Optimization and Machine Learning*, INFORMS Annual Meeting, Indianapolis, IN, USA
- 2018 – 2020 **Vice President**, *American Mathematical Society Graduate Student Chapter at NC State University*
- Organized the [Triangle Area Graduate Mathematics Conference for Spring 2019](#) [↗](#).
- Organized recruitment and social events for the NC State mathematical community.
- 2018 – 2020 **President**, *Undergrads Union Grads (UUG)*, Department of Mathematics, NC State University
Coordinated a mentoring program for undergraduate math majors with grad student advisors.
- 2018 – 2020 **Organizer**, *Nonlinear Analysis Graduate Student Workshop*, NC State University

REFEREEING

- 2026 **SIAM Journal on Optimization**
- 2026 **Operations Research Letters**
- 2026 **Conference on Neural Information Processing Systems (NeurIPS)**
- 2026 **Open Journal of Mathematical Optimization (OJMO)**

- 2026 **Journal of Optimization Theory and Applications (JOTA)**
- 2025 **Mathematical Programming**
- 2024 **Journal of Machine Learning Research (JMLR)**
- 2024 **Conference on Neural Information Processing Systems (NeurIPS)**
- 2024 **Mathematical Programming**
- 2023 **Journal of Machine Learning Research (JMLR)**
- 2023 **International Conference on Learning Representations (ICLR)**
- 2023 **Conference on Neural Information Processing Systems (NeurIPS)**
- 2023 **International Conference on Machine Learning (ICML)**
- 2023 **Advances in Computational Mathematics (ACOM)**
- 2022 **Conference on Neural Information Processing Systems (NeurIPS)**
- 2020 & 2021 **Journal of Mathematical Analysis and Applications (JMAA)**
- 2020 **Journal of Approximation Theory (JAT)**

SELECTED PRESENTATIONS

- LANS Seminar, Argonne National Lab.** Lemont, IL, USA. September, 2026
 - Invited lecture: *Recent results on nonconvex Frank-Wolfe splitting algorithms*
- SIAM Conference on Optimization 2026 (SIAM OP26).** Edinburgh, Scotland, UK. June, 2026
 - Invited minisymposium lecture: *Frank-Wolfe with Moreau envelope smoothing: a story about nonconvex splitting problems*
- Mathematical Foundations of Artificial Intelligence Seminar at Umeå University.** Umeå, Sweden (online), Sept., 2025
 - Invited Lecture: *Flexible block-iterative analysis for the Frank-Wolfe algorithm*
- International Conference on Continuous Optimization (ICCOPT).** Los Angeles, CA, USA. July, 2025
 - *Which is faster: Linear minimization or Projection?*
- INFORMS Annual Meeting.** Seattle, WA, USA. October, 2024
 - Invited Lecture: *Deterministic block-iterative analysis for the Frank-Wolfe algorithm*
- International Symposium on Mathematical Programming (ISMP).** Montreal, QC, CA. July 24, 2024
 - Invited Lecture: *Deterministic block-iterative analysis for the Frank-Wolfe algorithm* [🔗](#)
- INFORMS Annual Meeting.** Phoenix, AZ, USA. October 18, 2023
 - Invited Lecture: *Splitting the Conditional Gradient Algorithm* [🔗](#)
- Imaging Inverse Problems - Regularization, low dimensional models and applications**
(Mathématiques de l'Imagerie et de ses Applications CNRS workshop). Bordeaux, FR. March 23, 2023
 - *Signal recovery from inconsistent nonlinear observations* [🔗](#)
- MATH+ Spotlight Series.** Berlin, Germany. November 2, 2022
 - *A Frank-Wolfe approach for abs-smooth optimization*
- INFORMS Annual Meeting.** Indianapolis, IN, USA. October 16, 2022
 - Invited lecture: *A Frank-Wolfe approach for abs-smooth optimization*
- ZIB Tutorial Lecture Series.** Berlin, Germany. March 16, 2022
 - *Proximity operators and nonsmooth optimization* (online)
- European Signal Processing Conference (EUSIPCO) 2020.** Amsterdam, NL. January 21, 2021
 - *A fixed point framework for recovering signals from nonlinear transformations* (online)
- Nonlinear Analysis Graduate Student Workshop.** NC State, Raleigh, NC, USA. Fall 2020
 - *Signal recovery from nonlinear transformations*
- Triangle Area Graduate Math Conference.** UNC, Chapel Hill, NC, USA. Fall 2019
 - *The Gospel of Proximal Calculus: Optimization for non-differentiable problems*
- SIAM Graduate Student Tutorial.** NC State, Raleigh, NC, USA. March 21, 2019
 - *Proximal methods for optimization*
- Operator Splitting Workshop.** SAMSI, Research Triangle Park, NC, USA. March 21–23, 2018
 - Invited lecture: *Composite Infimal Convolutions*
- Applied Math Graduate Student Seminar** NCSU, Raleigh, NC, USA. Fall 2016
 - *Analyzing Multistationarity in Chemical Reaction Networks*
- Graduate Student Algebra Seminar** NCSU, Raleigh, NC, USA. Fall 2016
 - *Obstructions to Convexity and Neural Codes*
- Joint Mathematics Meetings** Seattle, WA, USA. January 6, 2016
 - *Analyzing Multistationarity in Chemical Reaction Networks*
- Texas A&M University Research Symposium** College Station, TX. July 23rd, 2015
 - *An Infinite Family of Networks with Non-Degenerate Equilibria* : Undergraduate lecture.
- M.A.A. MD-DC-VA Section Meeting** Salem, VA, USA. April 24–25, 2015
 - *Architectural Acoustic Oddities & The Asymptotic Behavior of Repetition Pitch*
- Acoustical Society of America Conference** Indianapolis, IN. Oct. 28, 2014
 - *Architectural Acoustic Oddities* (poster)
- Shenandoah Undergraduate Mathematics Conference (SUMS),** JMU, Harrisonburg, VA, USA.
 - 2024, 2018, 2016: Graduate School Panelist; 2015: Contributed talk; 2014: First place poster prize winner